

GEORGIOS KONTOGIANNIS

Ph.D. Candidate [ Google Scholar]

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RESEARCH INTERESTS

- Digital Signal Processing and Deep Learning for respiratory sound analysis.
- Deep generative modeling for synthesizing clinically meaningful audio sounds.
- Mitigating domain shift and heterogeneity in medical sound datasets.
- Trustworthy, robust, and deployment-aware machine learning in real-world healthcare systems.
- Graph-based information retrieval systems.

EDUCATION

University of Patras

Rio, Greece

Ph.D. Candidate, Department of Computer Engineering and Informatics
present

Sep 2023 –

- Advisor: Prof. Sotiris Nikolettseas
- Research focus: audio processing and algorithms development for respiratory sound analysis, multimodal AI modeling integrating digital and biological biomarkers, deep generative modeling for medical audio samples, robust and patient-centric diagnostic pipelines for edge-deployed clinical applications.

University of Patras

Rio, Greece

MSci., Department of Computer Engineering and Informatics

Sep 2018 – Jul 2023

- Advisor: Prof. Christos Makris
- Thesis: **A Study on Improvement of Graph-based Information Retrieval Models through Machine and Deep Learning Techniques**
- GPA: 7.81/10 — *Very Good* (Greek national scale: 5.0 pass, 8.5 Excellent, 10.0 maximum; equivalent to 78.1%)

ACADEMIC RESEARCH EXPERIENCE

CTI Diophantus — EN (POLLA) PLO Project, ML Researcher | Patras Feb 2026 – present

- Research contract with the Computer Technology Institute and Press “Diophantus” (CTI), developing ML/DL systems to optimize the nationwide school textbook distribution ecosystem under Greece’s new multiple-textbook policy.
- Development of predictive models for student population, textbook demand, and inventory forecasting; school clustering by textbook selection patterns; simulation of logistics and what-if supply-chain scenarios; integration of developed tools into the Publications Directorate IT infrastructure.
- Scientific Coordinator: Dr. Christos Manolopoulos

EU Horizon Research Project — SynAir-G, Research Scientist | University of Patras
Sep 2022 – present

- *Disrupting Noxious Synergies of Indoor Air Pollutants and their Impact in Childhood Health and Wellbeing, using Advanced Intelligent Multisensing and Green Interventions* (Grant Agreement No. 101057271).
- Investigation of synergistic interactions between indoor air quality metrics and physiological signals from wearable devices using multivariate and temporal causal models.
- Scientific Coordinator: Prof. Nikos Papadopoulos

INDUSTRIAL
RESEARCH
EXPERIENCE

Pfizer CDI — EyeAI Research Project, Research Scientist | University of Patras Jan 2024 – Oct 2025

- Externally funded Pfizer Collaborative Development Initiative (CDI) contract awarded to the University of Patras.
- Supervised dataset curation, annotation workflows, and quality control for ocular imaging.
- Developed an attention-gated UNet++ architecture for automatic sclera segmentation from smartphone-captured eye images.
- Quantized and deployed the model on-device for real-time smartphone inference (TensorFlow Lite / ONNX).

Pfizer CDI — Voice-Based Diagnostics (VBD-2), Research Scientist | University of Patras Jan 2023 – Sep 2024

- Designed and deployed smartphone-based cough detection and disease analysis pipeline for continuous passive respiratory screening, for real-time processing.
- Designed and deployed an end-to-end on-device AI pipeline processing 10-second tracheal and chest-wall auscultation recordings via the built-in microphone, with percussive noise removal and frequency-spectrum correction to mitigate domain shift across auscultation sites and microphone devices, for asthma and COPD detection.
- Developed a ResNet-34 classifier with a probabilistic voting mechanism for robust and explainable disease identification; developed deep generative models for cough and lung sound synthesis to address training data scarcity.

Pfizer CDI — Voice-Based Diagnostics (VBD-1), Research Scientist | University of Patras Jan 2022 – Aug 2023

- Developed cough detection algorithms and abnormal–normal lung sound classifiers deployed for real-time smartphone inference using the built-in microphone.

CLINICAL &
INSTITUTIONAL
COLLABORA-
TIONS

University General Hospital of Patras — Department of Internal Medicine, Research Scientist | Rio Nov 2025 – present

- Supervised dataset curation, annotation workflows, and quality control for respiratory infection data (audio and imaging).
- Multimodal deep-learning research combining paired lung sound recordings, chest X-rays, and inflammatory biomarkers for distinguishing bacterial vs. viral pneumonia and identifying clinically meaningful audio biomarkers.
- Development of mobile and web-based clinical data acquisition tools for in-hospital deployment.
- Clinical collaborator: Prof. Dr. Dimitris Velissaris

University General Hospital of Patras — Department of Internal Medicine, Research Scientist | Rio Jan 2024 – present

- Supervised curation, annotation, and quality control of ocular images for jaundice assessment.
- Statistical and causal modeling of digital biomarkers and hematological indicators in patients diagnosed with jaundice.
- Study conducted under institutional ethics committee approval (University General Hospital of Patras IRB).
- Clinical collaborator: Prof. Dr. Dimitris Velissaris

PUBLICATIONS REFEREED JOURNAL ARTICLES

- [C]: CONFERENCE
[J]: JOURNAL
[*]: FIRST AUTHOR
[†]: CORRESPONDING AUTHOR
[IP]: IN PRESS
[U]: UNDER REVIEW
- [J3] **G. Kontogiannis†***, P. Tzamalīs, S. Nikolettseas, T. Karamanidou, and T. G. Stavropoulos, “On-device Cough Detection and Respiratory Disease Classification Enhanced by Generative Data Augmentation,” *Computers in Biology and Medicine*, 2026. (IF: 6.3)
- [J2] N.-R. Kalogeropoulos†, **G. Kontogiannis***, and C. Makris, “Spectral clustering and query expansion using embeddings on the graph-based extension of the set-based information retrieval model,” *Expert Systems with Applications*, vol. 263, p. 125771, 2025. (IF: 7.5)
- [J1] P. Kapetanidis, F. Kalioras, C. Tsakonas, P. Tzamalīs†, **G. Kontogiannis**, T. Karamanidou, T. G. Stavropoulos, and S. Nikolettseas, “Respiratory diseases diagnosis using audio analysis and artificial intelligence: a systematic review,” *Sensors*, vol. 24, no. 4, p. 1173, 2024. (IF: 3.4)

REFEREED CONFERENCE PROCEEDINGS

- [C2] **G. Kontogiannis†***, P. Tzamalīs, and S. Nikolettseas, “Metadata-Conditioned Audio Transformers for Adaptive Respiratory Sound Classification,” in *Proc. 34th European Signal Processing Conference (EUSIPCO)*, [IP], 2026.
- [C1] **G. Kontogiannis†***, P. Tzamalīs, and S. Nikolettseas, “Exploring the Impact of Synthetic Data on Human Gesture Recognition Tasks Using GANs,” in *Proc. 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, pp. 384–391, IEEE, 2024.

UNDER REVIEW

- [U2] **G. Kontogiannis†***, P. Tzamalīs, D. Velissaris, and S. Nikolettseas, “AI-Based Jaundice Assessment from Ocular Images: A Structured Survey of Methods, Datasets, and Open Challenges,” *Under Review*, Healthcare Informatics Research (Springer Nature), 2026. (IF: 4.0)
- [U1] **G. Kontogiannis†***, P. Tzamalīs, S. Nikolettseas, T. Karamanidou, and T. G. Stavropoulos, “Transforming Smartphones into Stethoscopes: Edge AI for Asthma and COPD,” *Under Review*, IEEE Journal of Biomedical and Health Informatics (JBHI), 2026. (IF: 7.0)

GRANT WRITING EXPERIENCE

- **Pfizer CDI — Voice-Based Diagnostics (VBD-2)** 2023: Critically contributed to the research proposal (methodology, technical approach, and preliminary data) that secured the externally funded CDI contract with the University of Patras for smartphone-based respiratory disease monitoring.
- **Pfizer CDI — EyeAI Research Project** 2024: Critically contributed to the research proposal that secured the CDI contract for ocular biomarker analysis through computer vision methods, including system design and edge deployment strategy.

AWARDS & SCHOLARSHIPS

- University of Patras PhD Research Scholarship 2023 – 2025

TALKS & PRESENTATIONS

- **G. Kontogiannis†***, P. Tzamalīs, S. Nikolettseas, “Metadata-Conditioned Audio Transformers for Adaptive Respiratory Sound Classification,” *EUSIPCO 2026* 2026
- **G. Kontogiannis†***, P. Tzamalīs, S. Nikolettseas, “Exploring the Impact of Synthetic Data on Human Gesture Recognition Tasks Using GANs,” *DCOSS-IoT 2024* 2024

TEACHING EXPERIENCE	Guest Lecturer — Artificial Intelligence Algorithms and Applications for IoT Rio Spring 2026 ■ University of Patras, Department of Computer Engineering & Informatics. Course coordinator: Prof. Sotiris Nikolettseas. ■ Designed and delivered a six-lecture applied ML module for IoT time-series data, developing all materials from scratch: LaTeX slide decks with full mathematical derivations, Python demonstrations on real datasets, and a graded end-to-end project on the PAMAP2 activity recognition dataset. Digital Electronics Laboratory, Lab Instructor Computer Engineering & Informatics, University of Patras Feb 2022 – Jun 2022 ■ Instructed undergraduate students through hands-on digital circuit experiments: delivered in-lab theory sessions, guided experimental execution, and led result interpretation and critical analysis discussions. ■ Supervisor: Dr. Georgios Oikonomou Analog Electronics Laboratory, Lab Instructor Computer Engineering & Informatics, University of Patras Sep 2021 – Jan 2022 ■ Instructed undergraduate students through hands-on analog circuit experiments: delivered in-lab theory sessions, guided experimental execution, and led result interpretation and critical analysis discussions. ■ Supervisor: Dr. Georgios Oikonomou
	MENTORSHIP & LEADERSHIP ■ Alexandros Vallatos, MSc — <i>Causal Artificial Intelligence in IoT Health Data</i> ■ Miltiadis Mades, MSc — <i>Supervised Domain Adaptation Techniques for Classifying Pathological Respiratory Sounds</i> ■ Chrisavgi Pateli, MSc — <i>Methods and Techniques for the Automatic Detection and Segmentation of the Respiratory Cycle from Lung Sound Signals</i>
	SERVICE ■ Reviewer, IEEE Transactions on Biomedical Engineering (TBME): 2025 Server Manager — IoT Laboratory, Department of CEID University of Patras 2023 – present ■ Deployed and maintain the lab’s ML experimentation infrastructure: containerized MLflow tracking server with PostgreSQL and MinIO backends using Docker Compose, exposed via Nginx reverse proxy. ■ Administered Linux server environment including SSH access, user permissions, systemd services, and port forwarding; resolved multi-container networking and service binding issues. ■ Hosted and maintained backend services for multiple IoT Laboratory web platforms, including web server configuration, reverse proxy routing, and continuous uptime management.
TECHNICAL SKILLS	■ Programming & Scripting: Python, Java, C, C++, MATLAB, Bash ■ ML & Deep Learning: Scikit-Learn, TensorFlow, PyTorch, Ajax, Hugging Face Transformers, TensorFlow Lite, ONNX, MLflow, Weights & Biases ■ Signal Processing & Vision: librosa, OpenCV, NumPy, SciPy, Pandas, Matplotlib, Seaborn ■ DevOps & Infrastructure: Docker, Docker Compose, Nginx, Systemd, Linux Server Administration ■ Mobile Development: Flutter, Android SDK ■ Databases & Storage: MySQL, PostgreSQL, Cassandra, MongoDB, Elasticsearch, MinIO ■ Development Tools: LaTeX, Git, GitHub, Atlassian Suite
SPOKEN LANGUAGES	■ Greek — Native ■ English — Proficiency (Michigan ECPE)

REFERENCES

- Available upon request.